IMRS — Continuous Workflow After Step 07 (Frozen Anchor Weights)

Yansheng Luo

2026-03-03

Context: What Step 07 Produced and Why It Matters

After **Step 07**, we have a **frozen weight vector** (one weight per gene) derived **only from anchor datasets** via meta-analysis with heterogeneity control and information stabilization.

Conceptually:

- Anchor phase: learn a robust **direction in gene space** (the innate activation axis).
- Transfer phase (this document): score new datasets by projecting standardized expression onto that axis.
- Calibration datasets are used **only to evaluate** generalization (effect sizes / AUC), not to update weights.

The weights are permanently fixed after Step 07 and are never tuned again.

What IMRS Score Is (Definition)

For a target dataset D with control samples C, and frozen gene weights w_g:

1. Normalize counts within dataset D (DESeq2 size factors on all samples).
2. Transform expression:
$$x_{g,s} = \log_2(\text{normalized counts}_{g,s} + 1)$$
3. Standardize each gene against controls:
$$z_{g,s} = \frac{x_{g,s} - \mu_{g,C}}{\sigma_{g,C}}$$
4. Score each sample as a weighted sum:
$$\text{IMRS}_s = \sum_{g \in G} w_g \, z_{g,s}$$

This is a **frozen linear scoring function**, not a refit or optimized classifier.

Optionally, for reporting within a dataset, we can z-scale the final score using that dataset's controls:

$$\text{IMRSz}_s = \frac{\text{IMRS}_s - \mu_{\text{IMRS},C}}{\sigma_{\text{IMRS},C}}$$

This makes the score interpretable as "standard deviations above control baseline".

Step 08 — Score Computation on a Target Dataset

Inputs

- Raw integer counts: `03_counts/<DATASET>/.../gene_counts_clean.tsv`
- Design table containing:
  - `sample_id`
  - `condition_simple` in {CONTROL, DELIVERY}
- Frozen weights: `05_score/anchors/gene_weights.tsv`

Outputs

- `imrs_scores.tsv`:
  - sample_id
  - condition_simple
  - imrs_raw
  - imrs_z
  - coverage
- `qc_summary.tsv`

Design Decisions (Fixed Defaults)

- Minimum controls required: $n \geq 3$
- Gene SD floor: 0.10
- Final score SD floor: 0.10
- Minimum gene coverage: 0.80
- Strip Ensembl version suffix: TRUE

Implementation Logic

Step 08A — Normalize Counts

Use DESeq2 size-factor normalization on all samples with design `~ 1`.

Then compute:

$$x_{g,s} = \log_2(\text{norm}_{g,s} + 1)$$

Rationale: stabilizes variance and prevents dominance of extreme counts.

Step 08B — Control-Based Gene Standardization

For each gene:

$$\mu_{g,C} = \text{mean of controls}$$

$$\sigma_{g,C} = \text{sd of controls}$$

Apply SD floor:

$$\sigma^*_{g,C} = \max(\sigma_{g,C}, 0.10)$$

Then compute:

$$z_{g,s} = \frac{x_{g,s} - \mu_{g,C}}{\sigma^*_{g,C}}$$

If number of controls < 3, dataset fails scoring.

Step 08C — Apply Frozen Weights

Match genes between weights and expression matrix:

$$\text{coverage} = \frac{|G_{\text{overlap}}|}{|G_{\text{weights}}|}$$

Require coverage ≥ 0.80.

Then compute:

$$\text{IMRS}_s = \sum_{g \in G} w_g z_{g,s}$$

This is equivalent to a dot product between the sample expression vector (standardized) and the frozen weight vector.

Step 08C' — Optional Score Standardization

$$\mu_{\text{IMRS},C} = \text{mean of IMRS in controls}$$

$$\sigma^*_{\text{IMRS},C} = \max(\text{sd of controls}, 0.10)$$

$$\text{IMRSz}_s = \frac{\text{IMRS}_s - \mu_{\text{IMRS},C}}{\sigma^*_{\text{IMRS},C}}$$

QC Rules

1. Counts must be integer.
2. Controls ≥ 3.
3. Coverage ≥ 0.80.
4. Controls centered around 0 after IMRSz.
5. Report number of genes hitting SD floor.
6. Report top contributors: $w_g z_{g,s}$ ranked by magnitude.

Step 09 — Calibration Evaluation (No Retraining)

Calibration datasets are scored identically.

Report:

$$\Delta = \overline{\text{IMRSz}}_{\text{DELIVERY}} - \overline{\text{IMRSz}}_{\text{CONTROL}}$$

Optional:

- AUC (binary delivery vs control)
- Correlation with innate markers

Critical rule: weights are never modified during calibration.

Interpretation

- IMRS_raw: weighted projection of standardized expression onto innate activation axis.
- IMRS_z: standardized score within dataset; interpretable as SD above control.
- Anchors build weights.
- Calibration evaluates generalization.
- Future datasets are scored without retraining.

Conceptual Meaning

IMRS is not:

- A machine learning classifier
- A retrained regression
- A hyperparameter-optimized model

IMRS is:

- A frozen biological linear scoring function
- A projection onto a biologically defined innate-response direction
- Equivalent mathematically to a dot product in gene-expression space

It quantifies:

Weighted control-centered deviation of innate immune genes

Practical Use

IMRS allows:

- Ranking gene delivery systems by innate activation strength
- Comparing timepoints
- Cross-platform transfer (without retraining)
- Quantitative innate toxicity indexing

This completes the continuous workflow after Step 07.

Human Dataset Integration: Cross-Species Transfer Validation

Context and Constraint

We currently possess a single human RNA-seq dataset suitable for cross-sectional evaluation. Given that the IMRS weights were derived exclusively from mouse anchor datasets (Step 07), this human dataset **must not** be used for training, tuning, recalibration, or weight estimation.

Instead, it will be used strictly as an **external cross-species transfer test**.

The objective is:

> Does the mouse-derived innate activation axis detect innate immune activation in a human gene delivery dataset?

Principle of Transfer

Let:

- w_g^{mouse} = frozen mouse weights
- M = set of mouse core genes
- H = set of human orthologs of M

We define a mapping function:

$$\phi : M \to H$$

where $\phi(g)$ is a **one-to-one ortholog mapping**.

Only genes satisfying strict one-to-one orthology are retained.

Step H1: Ortholog Mapping

Mapping Rules

1. Use Ensembl one-to-one ortholog annotations.
2. Exclude:
   - one-to-many mappings
   - many-to-one mappings
   - genes without orthologs
3. Preserve mouse weight magnitude:

$$w_{\phi(g)}^{human} = w_g^{mouse}$$

Coverage Metric

$$\text{Ortholog Coverage} = \frac{|H|}{|M|}$$

This must be reported.

Low coverage reduces interpretability.

Step H2: Human Dataset Scoring

Scoring follows the exact Step 08 pipeline, but within the human dataset.

Normalization

Apply DESeq2 size-factor normalization across all samples with design ~ 1.

$$x_{g,s}^{human} = \log_2(\text{normalized counts}_{g,s} + 1)$$

Control-Based Standardization

For human controls:

$$z_{g,s}^{human} = \frac{x_{g,s}^{human} - \mu_{g,C}^{human}}{\sigma_{g,C}^{human}}$$

with SD floor applied as in mouse scoring.

Score Computation

$$\text{IMRS}_s^{human} = \sum_{g \in H} w_g^{human} \, z_{g,s}^{human}$$

Optional control-standardized score:

$$\text{IMRSz}_s^{human} = \frac{\text{IMRS}_s^{human} - \mu_C}{\sigma_C}$$

Step H3: Evaluation Metrics

Because only one human dataset is available, evaluation must remain conservative.

Primary Metric

$$\Delta = \overline{\text{IMRSz}}_{\text{DELIVERY}} - \overline{\text{IMRSz}}_{\text{CONTROL}}$$

Optional Metrics

- AUC (binary DELIVERY vs CONTROL classification)
- Distributional separation plots
- Top gene contribution inspection

Critical Restrictions

- Human data must **not** be used to modify weights.
- No hyperparameter tuning based on human results.
- No merging of human data into mouse anchor meta-analysis.

The human dataset serves exclusively as a **transfer validation dataset**.

Interpretation of Results

If:

$$\overline{\text{IMRSz}}_{\text{DELIVERY}} > \overline{\text{IMRSz}}_{\text{CONTROL}}$$

then the innate activation axis exhibits preliminary cross-species conservation.

If not, potential explanations include:

- Species-specific innate pathway divergence
- Incomplete ortholog mapping
- Platform differences
- Dataset-specific experimental factors

Scientific Positioning

With one human dataset, the correct claim is:

> We performed a cross-species transfer test by mapping mouse IMRS weights to human orthologs and scoring an independent human dataset. This provides preliminary evidence of conservation of the innate activation axis. Broader validation will require additional human datasets.

Future Extension

Once multiple human datasets become available:

1. Construct a human anchor model (parallel to mouse).
2. Perform cross-species weight correlation analysis:

$$\mathrm{cor}(w^{mouse}, w^{human})$$

3. Identify conserved innate genes.
4. Optionally construct a cross-species meta-axis.

Conclusion

Given the current constraint of a single human dataset, the optimal strategy is strict cross-species transfer validation using ortholog projection, without retraining or recalibration.

This preserves:

- Statistical defensibility
- Biological clarity
- Translational credibility